

ECE 3210 hw03

1. Compute the following convolutions.

(a) $y_1(t) = (u(t-1) - u(t-3)) * 2\text{rect}(\frac{t}{4})$

Solution: Let $x(t) = u(t-1) - u(t-3)$ and $h(t) = 2\text{rect}(\frac{t}{4})$. Then $y_1(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau) d\tau$.

The overlap of $[1, 3]$ with $[t-2, t+2]$ changes at $t = -1, 1, 3, 5$.

Case $t < -1$: the overlap is empty, so $y_1(t) = 0$.

Case $-1 \leq t < 1$: the overlap is $[1, t+2]$, so $y_1(t) = \int_1^{t+2} 2 d\tau = 2(t+1)$.

Case $1 \leq t < 3$: the pulse is fully inside the shifted rectangle, so $y_1(t) = \int_1^3 2 d\tau = 4$.

Case $3 \leq t \leq 5$: the overlap is $[t-2, 3]$, so $y_1(t) = \int_{t-2}^3 2 d\tau = -2t + 10$.

Case $t > 5$: the overlap is empty again, so $y_1(t) = 0$.

(b) $y_2(t) = (u(t-1) - u(t-3)) * \cos(\pi t)$

Solution: The convolution is $y_2(t) = \int_{-\infty}^{\infty} (u(\tau-1) - u(\tau-3)) \cos(\pi(t-\tau)) d\tau = \int_1^3 \cos(\pi(t-\tau)) d\tau$.

Since the same fixed limits apply for all t , $y_2(t) = \int_1^3 \cos(\pi(t-\tau)) d\tau = \frac{1}{\pi} [\sin(\pi(t-1)) - \sin(\pi(t-3))] = 0$.

2. Compute the following convolutions.

(a) $y_1(t) = \sin(\pi t)u(t+1) * u(t-2)$

Solution: Since $u(t-2)$ requires the flipped signal to overlap $\tau \leq t-2$, $y_1(t) = \int_{-\infty}^{\infty} \sin(\pi\tau)u(\tau+1)u(t-\tau-2) d\tau$.

Case $t < 1$: the overlap is empty, so $y_1(t) = 0$.

Case $t \geq 1$: the overlap is $[-1, t-2]$, so $y_1(t) = \int_{-1}^{t-2} \sin(\pi\tau) d\tau = -\frac{\cos(\pi t)+1}{\pi}$.

(b) $y_2(t) = 2(u(t+1) - u(t-1)) * e^{-t}(u(t-1) - u(t-4))$

Solution: Write the convolution as $y_2(t) = \int_{-\infty}^{\infty} 2(u(\tau+1)-u(\tau-1))e^{-(t-\tau)}(u(t-\tau-1) - u(t-\tau-4)) d\tau$.

The overlap interval is $[-1, 1] \cap [t-4, t-1]$.

Case $t < 0$: the overlap is empty, so $y_2(t) = 0$.

Case $0 \leq t < 2$: the overlap is $[-1, t-1]$, so $y_2(t) = \int_{-1}^{t-1} 2e^{-(t-\tau)} d\tau = 2e^{-1}(1 - e^{-t})$.

Case $2 \leq t < 3$: the overlap is the full interval $[-1, 1]$, so $y_2(t) = \int_{-1}^1 2e^{-(t-\tau)} d\tau = 2e^{-t}(e - e^{-1})$.

Case $3 \leq t < 5$: the overlap is $[t-4, 1]$, so $y_2(t) = \int_{t-4}^1 2e^{-(t-\tau)} d\tau = 2(e^{1-t} - e^{-4})$.

Case $t \geq 5$: the overlap is empty again, so $y_2(t) = 0$.

3. Compute the following convolutions.

(a) $y_1(t) = \sin(t)(u(t) - u(t - 2\pi)) * u(t)$

Solution: Convolving with $u(t)$ accumulates the area under $\sin(t)$ over the window $[0, 2\pi]$. Write $y_1(t) = \int_{-\infty}^{\infty} \sin(\tau)(u(\tau) - u(\tau - 2\pi))u(t - \tau) d\tau$.

Case $t < 0$: the overlap is empty, so $y_1(t) = 0$.

Case $0 \leq t \leq 2\pi$: the overlap is $[0, t]$, so $y_1(t) = \int_0^t \sin(\tau) d\tau = 1 - \cos(t)$.

Case $t > 2\pi$: the overlap is the full window $[0, 2\pi]$, so $y_1(t) = \int_0^{2\pi} \sin(\tau) d\tau = 0$.

(b) $y_2(t) = u(t) * t(u(t + 1) - u(t - 2))$

Solution: The convolution is $y_2(t) = \int_{-\infty}^{\infty} u(\tau)(t - \tau)(u(t - \tau + 1) - u(t - \tau - 2)) d\tau$.

The overlap of the ramp with the step begins at $t = -1$ and finishes at $t = 2$.

Case $t < -1$: the overlap is empty, so $y_2(t) = 0$.

Case $-1 \leq t < 2$: the overlap is $[0, t]$, so $y_2(t) = \int_0^t \tau d\tau = \frac{t^2}{2} - \frac{1}{2}$.

Case $t \geq 2$: the overlap is the full support $[0, 2]$, so $y_2(t) = \int_0^2 \tau d\tau = \frac{3}{2}$.

4. Compute the following convolutions.

(a) $y_1(t) = e^{-t}u(t) * (u(t-1) - u(t-3))$

Solution: The shifted rectangular pulse creates two overlap cases. Write $y_1(t) = \int_{-\infty}^{\infty} e^{-\tau}u(\tau)(u(t-\tau-1) - u(t-\tau-3)) d\tau$.

Case $t < 1$: the overlap is empty, so $y_1(t) = 0$.

Case $1 \leq t < 3$: the overlap is $[1, t]$, so $y_1(t) = \int_1^t e^{-(t-\tau)} d\tau = 1 - e^{-(t-1)}$.

Case $t \geq 3$: the overlap is the full pulse interval $[1, 3]$, so $y_1(t) = \int_1^3 e^{-(t-\tau)} d\tau = e^{-t}(e^3 - e^1)$.

(b) $y_2(t) = (u(t-1) - u(t-3)) * \cos(\pi t)$

Solution: The rectangle has width two, so the cosine integrates over exactly one period and cancels. In integral form, $y_2(t) = \int_1^3 \cos(\pi(t-\tau)) d\tau = 0$.

5. Compute the following convolutions.

(a) $y(t) = 2u(t) * t(u(t) - u(t-1))$

Solution: Write the convolution as $y(t) = \int_{-\infty}^{\infty} 2u(\tau) \tau(u(t-\tau) - u(t-\tau-1)) d\tau$.

Case $t < 0$: the overlap is empty, so $y(t) = 0$.

Case $0 \leq t < 1$: the overlap is $[0, t]$, so $y(t) = \int_0^t 2\tau d\tau = t^2$.

Case $t \geq 1$: the overlap is the full support $[0, 1]$, so $y(t) = \int_0^1 2\tau d\tau = 1$.

(b) $y(t) = \text{rect}\left(\frac{t}{4}\right) * \sin(\pi t) \text{rect}\left(\frac{t}{2}\right)$

Solution: Let $y(t) = \int_{-\infty}^{\infty} \text{rect}\left(\frac{\tau}{4}\right) \sin(\pi(t-\tau)) \text{rect}\left(\frac{t-\tau}{2}\right) d\tau$.

The overlap interval has three regimes: left overlap, centered cancellation, and right overlap.

Case $t < -3$: the overlap is empty, so $y(t) = 0$.

Case $-3 \leq t \leq -1$: the overlap is $[-2, t+1]$, so $y(t) = \int_{-2}^{t+1} \sin(\pi(t-\tau)) d\tau = -\frac{\cos(\pi t) + 1}{\pi}$.

Case $-1 < t < 1$: the overlap is the full window $[t-1, t+1]$, and the integral cancels, so $y(t) = \int_{t-1}^{t+1} \sin(\pi(t-\tau)) d\tau = 0$.

Case $1 \leq t \leq 3$: the overlap is $[t-1, 2]$, so $y(t) = \int_{t-1}^2 \sin(\pi(t-\tau)) d\tau = \frac{1 + \cos(\pi t)}{\pi}$.

Case $t > 3$: the overlap is empty again, so $y(t) = 0$.

6. Compute the following convolution.

(a) $y_1(t) = e^{-2t}u(t) * e^{-t}u(t)$

Solution: Both signals are causal, so the overlap starts at $t = 0$. Write $y_1(t) = \int_{-\infty}^{\infty} e^{-2\tau}u(\tau)e^{-(t-\tau)}u(t-\tau) d\tau$.

Case $t < 0$: the overlap is empty, so $y_1(t) = 0$.

Case $t \geq 0$: the overlap is $[0, t]$, so $y_1(t) = \int_0^t e^{-2\tau}e^{-(t-\tau)} d\tau = e^{-t} - e^{-2t}$.

7. Determine whether each system is stable, marginally stable, or unstable.

(a)

$$(D^4 + 6D^3 + 9D^2)y(t) = (D - 1)f(t)$$

Solution: The transfer function has a repeated pole at the origin, so the system is unstable.

(b)

$$h(t) = e^{-\frac{t}{2}}u(t)$$

Solution: The impulse response decays exponentially and is absolutely integrable, so the system is stable.

(c)

$$(D^3 + 10D^2 + 26D)y(t) = D^2f(t)$$

Solution: This system is marginally stable.

8. Determine whether each system is stable, marginally stable, or unstable.

(a)

$$(D^3 + 25D)y(t) = (D^2 + 5)f(t)$$

Solution: This system is marginally stable.

(b)

$$h(t) = e^{\frac{t}{2}}(u(t) - u(t - 10))$$

Solution: The impulse response has finite support, so it is absolutely integrable and the system is stable.

(c)

$$(D^6 + 10D^5 + 49D^4 + 110D^3 + 94D^2 - 120D - 144)y(t) = D^2f(t)$$

Solution: This system is unstable.